

Insider Cluster Alpha Study

Do clustered insider open-market purchases predict positive abnormal returns in U.S. equities?

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Abstract

This paper tests whether clustered insider open-market purchases predict positive abnormal stock returns in U.S. equities. I define a cluster event as 3 or more unique corporate insiders filing open-market purchases for the same stock within a rolling 30-calendar-day window, with the event date set to the last filing date in the cluster. All methodological choices — signal definition, universe filters, return construction, and statistical tests — are pre-registered before analysis. Applying this framework to SEC Form 4 filings from 2015 to 2024, I identify 2,664 cluster events across 1,030 unique firms. At the pre-registered primary horizon of 30 trading days, mean cumulative abnormal return (market-adjusted) is +1.65% ($N = 2,565$; $t = 5.42$, $p < 1e-7$), exceeding the Bonferroni-adjusted significance threshold. Abnormal returns are concentrated in the first trading day (+1.22%), consistent with rapid market incorporation of newly disclosed insider information. The result is robust to CAPM risk adjustment (30-day CAR = +2.99%; $t = 9.74$), alternative signal calibrations (2+ to 5+ insiders, 14- to 60-day windows), and subsample splits by time period and market capitalization. The findings suggest that temporal clustering of insider purchases conveys incremental information beyond what individual insider trades reveal, consistent with cluster events filtering for high-conviction, opportunistic insider behavior.

1. Introduction

Insider trading disclosures are among the most widely tracked corporate signals because insiders may possess material information about firm fundamentals not fully reflected in market prices. A large body of evidence confirms that insider purchases, on average, precede positive abnormal returns. However, most of this literature treats each insider transaction independently or aggregates at the firm-month level. An open question is whether coordinated or clustered buying — multiple distinct insiders purchasing the same stock within a short time window — conveys stronger information than isolated transactions.

This study asks a specific question: do clustered insider open-market purchases (3+ unique insiders in a 30-calendar-day window for the same ticker, with the event date defined as the last SEC filing date in the cluster) predict positive abnormal returns in U.S. equities?

I pre-registered all primary methodological choices before conducting the full-sample analysis. The pre-registration locks the signal definition, universe constraints (transaction price $\geq \$5$, market cap $\geq \$300M$), overlap handling (60-day same-ticker dedup), return construction (compounded stock minus compounded SPY), event-day convention (first trading day on or after the filing date), and the primary statistical test (two-sided t-test with Bonferroni threshold 0.0125 across four horizons). Pre-registration eliminates researcher degrees of freedom and ensures that the reported results constitute a confirmatory test under a pre-registered specification.

The primary finding is a positive and statistically significant mean abnormal return of +1.65% over the 30-trading-day window following the event date ($t = 5.42$, $p = 6.6e-08$). The effect is present across both halves of the sample period (2015–2019 and 2020–2024), survives CAPM risk adjustment, and persists under alternative signal calibrations.

This study makes three contributions to the insider trading literature. First, it defines and tests a specific cluster purchase signal — 3 or more unique insiders buying within a 30-day window — not previously examined as a distinct predictor of abnormal returns. Second, it employs a fully pre-registered design that eliminates specification search and ensures confirmatory rather than exploratory inference. Third, it provides comprehensive robustness evidence across CAPM risk adjustment, alternative signal calibrations, and temporal and size-based subsample splits.

The remainder of the paper proceeds as follows. Section 2 reviews related literature. Section 3 describes data sources and sample construction. Section 4 details the pre-registered methodology. Section 5 presents results. Section 6 reports robustness checks. Section 7 discusses economic interpretation and limitations. Section 8 concludes.

2. Literature Review

The informational content of insider trading has been studied since at least Jaffe (1974), who documents abnormal returns following months with heavy insider buying. Seyhun (1986) provides systematic evidence that insiders earn abnormal profits on their trades and that aggregate insider trading activity predicts market-wide returns. Seyhun (1998) extends this work in a comprehensive treatment of insider trading patterns and their regulatory implications.

Lakonishok and Lee (2001) confirm that insider purchases contain predictive information for future stock returns, though the signal is noisy at the individual transaction level. Their analysis suggests that aggregation across insiders and across time may improve signal quality.

Jeng, Metrick, and Zeckhauser (2003) construct calendar-time portfolios mimicking insider purchases and estimate abnormal returns of approximately 6% per year, providing portfolio-level evidence that the insider purchase signal survives standard risk adjustments.

Cohen, Malloy, and Pomorski (2012) introduce a distinction between routine and opportunistic insider trades, showing that opportunistic trades are far more informative. Their work highlights that not all insider transactions are equal and that filtering or weighting by trade characteristics can sharpen the signal. This motivates the present study's focus on cluster events, where multiple insiders act within a narrow window, as a potential marker of shared opportunistic conviction.

The regulatory context is relevant to event timing. Under SEC Section 16(a), corporate insiders must report transactions on Form 4 within two business days of the trade date. The filing date is therefore typically close to but not identical to the trade date. This study uses the filing date as the event date (pre-registered) because it represents the date at which the information becomes publicly available to outside investors. Any delay between trade date and filing date works against finding abnormal returns from the filing date forward, making the test conservative.

A gap in the existing literature is the systematic evaluation of clustered insider purchases as a distinct signal. While Lakonishok and Lee (2001) aggregate insider activity at the firm level and Cohen, Malloy, and Pomorski (2012) classify individual trades, no prior study to my knowledge defines and tests a specific

cluster rule — multiple unique insiders purchasing within a fixed calendar window — using a pre-registered methodology. This paper fills that gap by formally defining a cluster rule and evaluating its predictive power using a pre-registered event-study framework. While prior work shows that insider purchases contain predictive information, the literature has largely focused on individual trades or firm-level aggregation. This study instead examines whether temporal clustering of purchases by multiple insiders conveys incremental information.

3. Data

Insider filing data was sourced from OpenInsider, which provides structured records of SEC Form 4 filings. I collected all filings classified as open-market purchases from January 2015 through December 2024. The raw dataset contained 92,396 filing records. After standardizing ticker symbols, parsing dates, filtering to open-market purchases only, applying a \$5 minimum transaction price floor, and removing duplicate filings (same ticker, same date, same insider name), the cleaned dataset contained 90,424 filings across 5,696 unique tickers.

Daily adjusted closing prices were downloaded from Yahoo Finance via the yfinance Python library for all tickers present in the cleaned filings, with a start date of June 2014 (to support pre-event estimation windows) and an end date of January 2025. The resulting price panel covers 2,665 trading days across 3,441 tickers. Of the 5,696 unique tickers in the cleaned filings, 2,255 (39.6%) could not be matched to downloadable price data and were excluded. These failures include delisted firms, renamed tickers, and data provider gaps. This is a known source of potential survivorship bias and is discussed in Section 7. SPY adjusted close prices were downloaded separately over the same period.

Market capitalizations were obtained from Yahoo Finance. These values reflect the most recent available figure at the time of data collection rather than the value at the time of each event, and therefore constitute a look-ahead measure. This bias in the universe filter is documented as a limitation in Section 7.

Table 1: Sample Construction

Step	Count
Raw Form 4 filings (2015–2024)	92,396
After cleaning + price filter + dedup	90,424
Unique tickers in cleaned filings	5,696
Tickers with price data available	3,441
Cluster events detected (pre-filter)	6,490
After market cap $\geq$ \$300M filter	3,002
After 60-day overlap deduplication (final)	2,664
Unique tickers in final sample	1,030

Notes: Cleaning includes ticker standardization, date parsing, open-market purchase filter, transaction price $\geq$ \$5 floor, and duplicate removal (same ticker, date, and insider name). Cluster detection requires ≥ 3 unique insiders with filings within a 30-calendar-day window. Overlap deduplication retains only the first event when multiple clusters for the same ticker occur within 60 calendar days.

After applying the cluster detection algorithm (Section 4), universe filters (transaction price $\geq \$5$, market cap $\geq \$300\text{M}$), and 60-day overlap deduplication, the final sample contains 2,664 cluster events across 1,030 unique firms, spanning 10 years (approximately 266 events per year). Event frequency is elevated in 2020, consistent with broad insider buying during the COVID-19 market disruption. The median market capitalization of event firms is $\$1.57$ billion.

4. Methodology

All methodological choices described in this section are fixed by pre-registration and were locked before the primary analysis was conducted. This study was pre-registered on 2026-03-03 (see PREREGISTRATION.md).

Signal detection. For each ticker, I scan all cleaned filings chronologically using a greedy sliding-window algorithm. Starting from the earliest filing, I define a 30-calendar-day forward window and count the number of unique insider names with at least one open-market purchase filing in that window. If the count reaches 3 or more, a cluster event is recorded. The event date is set to the latest (most recent) filing date among the insiders in that cluster window. The scan then advances past the end of the current window and resumes. Clusters may therefore contain more than three insiders if additional filings occur within the same window. Tickers with fewer than 3 total unique insiders across the full sample are skipped entirely.

Universe filters. Raw cluster events are filtered to stocks with a transaction price of at least $\$5$ (applied during data cleaning) and a market capitalization of at least $\$300$ million (applied using the market cap data described in Section 3).

Overlap deduplication. For each ticker, if multiple cluster events occur within 60 calendar days of each other, only the first event is retained. This prevents the same underlying information episode from generating multiple correlated observations. The 60-day deduplication rule also ensures that cluster events for the same ticker do not overlap in their evaluation windows.

Event-day alignment. Day 0 is defined as the first trading day on or after the event date. If the event date (filing date) falls on a weekend or holiday, Day 0 is the next available trading day.

Abnormal return construction. For each event, cumulative abnormal return over window $[0, W]$ is computed as:

$$\text{CAR}(0, W) = \prod_{t=0}^W (1 + r_{\text{stock},t}) - 1 - \left[\prod_{t=0}^W (1 + r_{\text{SPY},t}) - 1 \right]$$

where $r_{\text{stock},t}$ and $r_{\text{SPY},t}$ are daily returns on trading day t . This covers $W + 1$ trading days (days 0 through W inclusive) because the event window includes day 0. If any daily return in the window is missing (NaN), the event's CAR for that window is set to NaN and excluded from the statistical test to avoid partial-window bias. Four horizons are evaluated: $W \in \{1, 5, 30, 60\}$. The primary pre-registered horizon is $\text{CAR}(0, 30)$.

This market-adjusted return approach follows standard event-study practice as described in MacKinlay (1997). SPY is used as the market benchmark because it provides a liquid proxy for the U.S. equity market and is widely used in event-study research.

Statistical test. For each horizon, I compute the cross-sectional mean CAR and test whether it differs from zero using a two-sided t-test. Because four horizons are tested, the significance threshold is Bonferroni-

adjusted to $0.05 / 4 = 0.0125$. The primary hypothesis test is the t-test for CAR(0, 30).

5. Results

The primary market-adjusted results across all four pre-registered horizons are presented in Table 2. The number of events decreases slightly at longer horizons because events with incomplete return windows are excluded.

Table 2: Market-Adjusted Cumulative Abnormal Returns by Horizon

Window	N	Mean	Median	Std	t-stat	p-value	Hit Rate	Significant
CAR(0,1)	2,603	+1.22%	+0.33%	6.06%	10.26	3.0e-24	60.6%	Yes
CAR(0,5)	2,603	+1.47%	+0.53%	8.14%	9.20	7.0e-20	55.1%	Yes
CAR(0,30)	2,565	+1.65%	+0.41%	15.42%	5.42	6.6e-08	51.8%	Yes
CAR(0,60)	2,553	+3.23%	+0.72%	26.82%	6.05	1.7e-09	53.1%	Yes

Notes: Significance assessed at Bonferroni-adjusted threshold of 0.0125 (= 0.05/4). Hit rate is the percentage of events with CAR > 0. N varies across horizons because events with missing daily returns within the evaluation window are excluded from that horizon's test.

All four horizons are statistically significant at the Bonferroni-adjusted threshold of 0.0125. The primary pre-registered test — CAR(0, 30) — shows a mean abnormal return of +1.65% with a t-statistic of 5.42 and p-value of 6.6e-08.

Several features of the results merit comment. First, abnormal returns are concentrated in the first few days after the event: the mean CAR(0, 1) of +1.22% represents the majority of the CAR(0, 5) of +1.47%. This pattern is consistent with rapid market incorporation of newly disclosed insider information, where the market responds quickly once the last cluster filing becomes public. Second, the gap between mean and median widens at longer horizons, indicating right-skew in the CAR distribution — a minority of events generate large positive returns that contribute disproportionately to the mean. Third, hit rates decline from 60.6% at the 1-day horizon to 51.8% at 30 days, suggesting that while the average effect is positive, individual event outcomes become noisier over longer windows.

Figures

Figure 1 plots the mean CAR across the four evaluation horizons with 95% confidence intervals. The steep rise between day 0 and day 1 (+1.22%) accounts for the bulk of the short-term effect. Returns continue to drift upward through day 60, but the confidence band widens substantially, reflecting increasing cross-sectional dispersion at longer horizons.

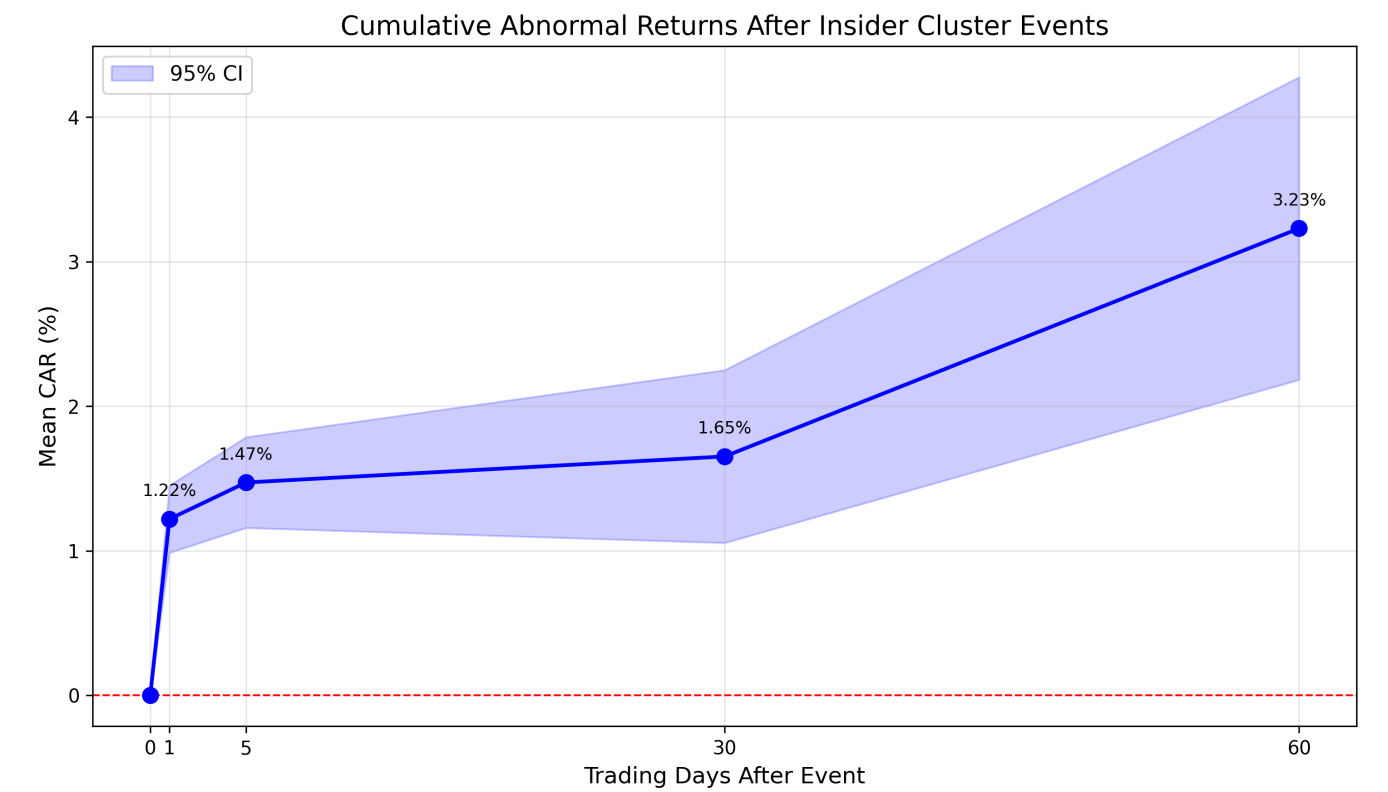


Figure 1: Mean cumulative abnormal return at 1, 5, 30, and 60 trading-day horizons. Shaded area represents the 95% confidence interval around the cross-sectional mean. Dashed line at zero represents no abnormal return.

Figure 2 displays the cross-sectional distribution of CAR(0, 30) across all 2,565 events with valid 30-day returns. The distribution is approximately centered near zero with a pronounced right tail. The median (+0.41%) lies well below the mean (+1.65%), confirming that a subset of large positive outcomes drives the mean effect. Despite this skew, 51.8% of events produce positive abnormal returns, indicating that the signal is not solely driven by outliers.

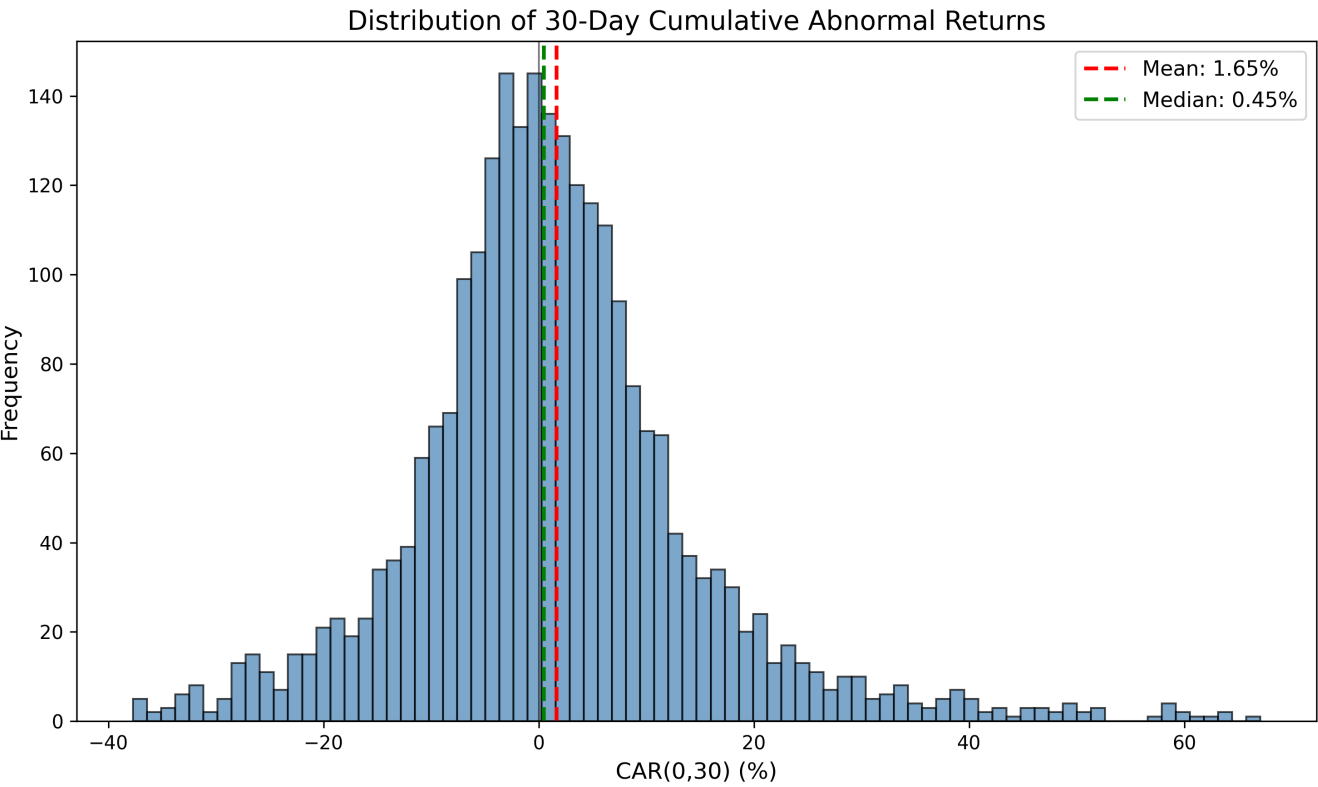


Figure 2: Histogram of 30-trading-day cumulative abnormal returns across 2,565 cluster events. Vertical solid line indicates the mean (+1.65%); vertical dashed line indicates the median (+0.41%).

Figure 3 shows the mean CAR(0, 30) by calendar year. The signal is positive in the majority of individual years, with no single year dominating the full-sample result. Elevated mean CARs in certain years, particularly 2020, are consistent with insiders purchasing during periods of market dislocation when private information about firm recovery prospects may have been particularly valuable.

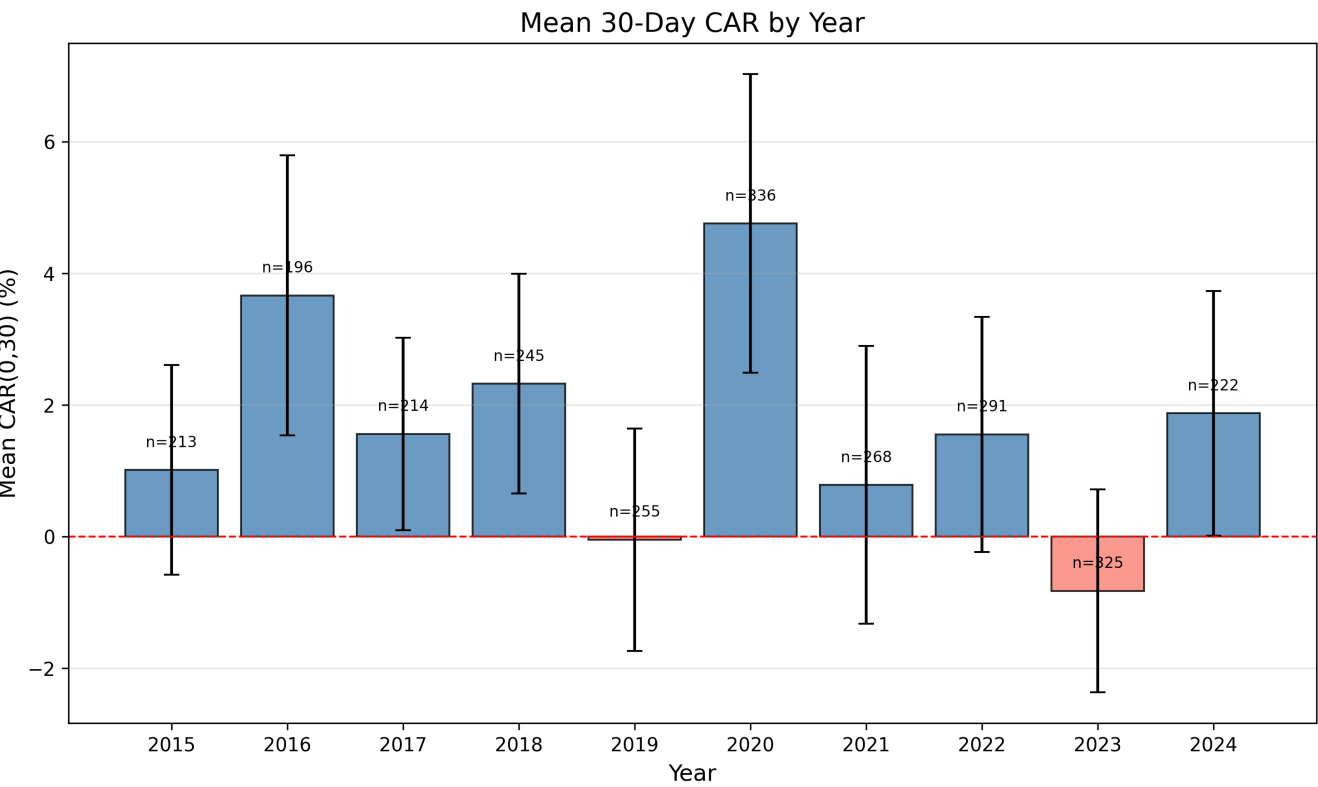


Figure 3: Mean 30-trading-day cumulative abnormal return by event year, 2015–2024. Error bars represent 95% confidence intervals. Dashed line at zero represents no abnormal return.

6. Robustness

Three categories of robustness checks are conducted.

6.1 CAPM-Adjusted Abnormal Returns

To address the concern that market-adjusted returns may reflect systematic risk exposure rather than genuine alpha, I estimate a CAPM model for each event using an estimation window of trading days $[-250, -30]$ relative to day 0. Daily stock returns are regressed on daily SPY returns to obtain alpha and beta estimates. During the event window, expected return on each day is $\hat{\alpha} + \hat{\beta} \times r_{\text{SPY},t}$, and the daily abnormal return is actual minus expected. Daily abnormal returns are summed rather than compounded because expected returns are estimated separately for each day. This additive aggregation follows standard event-study convention for risk-adjusted CARs and differs from the compounded construction used in the primary market-adjusted analysis. Events without sufficient return history to populate the estimation window are excluded from the CAPM analysis.

Table 3: CAPM-Adjusted Cumulative Abnormal Returns by Horizon

Window	N	Mean	Median	Std	t-stat	p-value	Significant
CAR(0,1)	2,273	+1.29%	+0.67%	5.94%	10.38	1.05e-24	Yes
CAR(0,5)	2,273	+1.82%	+0.74%	7.99%	10.85	8.92e-27	Yes
CAR(0,30)	2,235	+2.99%	+2.05%	14.52%	9.74	5.72e-22	Yes
CAR(0,60)	2,223	+5.74%	+4.05%	22.19%	12.20	3.48e-33	Yes

Notes: CAPM betas estimated using trading days $[-250, -30]$ relative to event day 0. Daily abnormal returns are summed (not compounded) over the event window. Events without sufficient estimation-window return history are excluded. Significance assessed at Bonferroni-adjusted threshold of 0.0125.

The CAPM-adjusted results are qualitatively consistent with the primary market-adjusted findings. At the 30-day horizon, mean CAPM-adjusted CAR is +2.99% ($t = 9.74$, $p = 5.72e-22$), confirming that the abnormal return is not explained by systematic market risk exposure alone. CAPM-adjusted point estimates are slightly larger than market-adjusted estimates across all horizons, which may reflect the removal of beta-related return attribution.

6.2 Signal Variant Sensitivity

I vary the cluster signal parameters to test whether results depend on the specific pre-registered calibration. Five variants are tested, as shown in Table 4.

Table 4: CAR(0, 30) Across Signal Variants

Variant	Insiders	Window	N Events	Mean CAR(0,30)	t-stat	p-value
Baseline (pre-reg)	3+	30 days	2,664	+1.65%	5.42	6.6e-08

Variant	Insiders	Window	N Events	Mean CAR(0,30)	t-stat	p-value
Loose threshold	2+	30 days	5,098	+1.63%	7.74	1.19e-14
Strict threshold	5+	30 days	1,024	+1.65%	2.30	2.18e-02
Short window	3+	14 days	2,359	+1.55%	4.98	6.81e-07
Long window	3+	60 days	2,735	+2.46%	6.81	1.23e-11

Notes: All variants use identical universe filters (transaction price $\geq$ |\$5, market cap $\geq$ |\$300M), 60-day overlap deduplication, and market-adjusted return computation. Only the signal detection parameters vary. Baseline row is the pre-registered specification. Statistical significance evaluated at the Bonferroni-adjusted threshold of 0.0125.

The signal is robust to moderate variation in calibration. Mean CARs are broadly similar across four of the five specifications. The strict threshold (5+ insiders) produces a comparable point estimate (+1.65%) but does not reach the Bonferroni-adjusted significance threshold ($p = 0.022$ vs. threshold of 0.0125), likely reflecting reduced statistical power due to the smaller sample size ($N = 1,024$). In contrast, the loose threshold (2+ insiders) with nearly double the event count produces highly significant abnormal returns (+1.63%, $p = 1.2e-14$), suggesting that the core phenomenon — multiple insiders purchasing within a concentrated window — remains informative across a range of definitions.

6.3 Subsample Stability

The full sample is split into two non-overlapping time periods (2015–2019, $N = 1,176$; 2020–2024, $N = 1,488$) and a large-cap subsample (top 75% by market cap, $N = 1,998$).

Table 5: CAR(0, 30) Across Subsamples

Subsample	N	Mean CAR(0,30)	t-stat	p-value
Full sample	2,565	+1.65%	5.42	6.6e-08
2015–2019	1,176	+1.63%	4.15	3.5e-05
2020–2024	1,488	+1.67%	3.73	2.0e-04
Large-cap (top 75%)	1,998	+1.95%	5.70	1.4e-08

Notes: Time-period subsamples are non-overlapping. Large-cap subsample includes events at or above the 25th percentile of market capitalization (i.e., excludes the smallest quartile). All subsamples use identical return computation and statistical testing. N reflects events with valid 30-day return windows; some events are excluded due to missing price data.

The primary CAR(0, 30) is statistically significant in both time periods (+1.63%, $p = 3.5e-05$ for 2015–2019; +1.67%, $p = 2.0e-04$ for 2020–2024) and in the large-cap subsample (+1.95%, $p = 1.4e-08$). The effect is not driven by a single subperiod or by small-cap effects. Event counts in the subsamples differ slightly from the full-sample N due to missing return data in the evaluation window.

7. Discussion and Limitations

Discussion

The primary result of this study — a mean 30-day cumulative abnormal return of +1.65% following clustered insider purchases — warrants discussion along four dimensions: economic magnitude, mechanism, tradability, and signal decay.

Economic magnitude. A +1.65% abnormal return over 30 trading days is economically meaningful at the individual event level. Naively annualized, this corresponds to approximately $1.65\% \times (252 / 30) \approx 13.9\%$ per year if an investor could continuously deploy capital into cluster events. However, this extrapolation overstates implementable returns because cluster events occur only approximately 266 times per year across 1,030 unique firms, and not all events can be traded simultaneously without portfolio concentration risk. For context, Jeng, Metrick, and Zeckhauser (2003) estimate abnormal returns of approximately 6% per year from calendar-time portfolios mimicking all insider purchases. The substantially higher per-event magnitude documented here is consistent with cluster events representing a filtered, higher-conviction subset of the broader insider purchase universe. Requiring 3 or more insiders to buy within a 30-day window selects for situations where private information is sufficiently compelling that multiple individuals act on it independently, raising the average signal quality relative to isolated individual insider transactions.

Mechanism. Why might clustered insider purchases be more informative than individual trades? Several non-exclusive explanations are consistent with the data. First, multiple insiders may independently observe favorable firm fundamentals — such as an improving order pipeline, cost reductions, or resolution of a pending uncertainty — and each conclude that the stock is undervalued. The temporal clustering of their purchases reflects convergent private assessments rather than coordinated action. Second, the cluster pattern may proxy for conviction concentration: when 3 or more insiders are willing to commit personal capital within a narrow window, the implied strength of the shared information signal is higher than when a single insider buys. This interpretation aligns with Cohen, Malloy, and Pomorski (2012), who show that opportunistic insider trades — those driven by information rather than routine portfolio rebalancing — are far more predictive of future returns. Cluster events may naturally filter toward opportunistic trades, since routine purchases by multiple insiders within the same month would be coincidental rather than systematic. It is important to note what cluster events are not: they do not necessarily indicate collusion, illegal coordination, or even mutual awareness among the insiders. Each insider files independently with the SEC, and the cluster pattern may emerge from independent reactions to the same underlying fundamentals.

Tradability. The sample contains 2,664 events over 10 years, or approximately 266 events per year — roughly one per trading day on average. The event date is the SEC filing date, which is publicly observable in real time via the EDGAR filing system. A trader monitoring Form 4 filings could identify cluster completions on the day they occur: when a third unique insider files a purchase for a given stock within 30 days of two prior filings, the cluster signal activates. However, the practical implementability of the signal is constrained by two factors. First, the $CAR(0, 1)$ of +1.22% indicates that the majority of the short-horizon abnormal return is captured on day 0 itself (the filing date or next trading day), implying that execution speed is critical and that delayed entry would substantially reduce realized returns. Second, no transaction cost modeling is performed in this study. Round-trip trading costs, bid-ask spreads, and market impact — particularly for the smaller-cap stocks in the sample — could erode a meaningful portion of the gross abnormal return. Whether the signal supports a net-profitable trading strategy remains an open question for future research.

Signal decay. The time-path of abnormal returns provides additional insight into the information incorporation process. Of the +1.65% mean CAR at 30 days, +1.22% is realized by the end of day 1, and

+1.47% by the end of day 5. This front-loading suggests that the market responds rapidly to the public disclosure of insider filings, with the bulk of the price adjustment occurring within the first trading session. The continued drift from day 5 to day 60 (+3.23% total CAR) may reflect slower diffusion of the underlying information to less attentive market participants, gradual fundamental realization (e.g., subsequent earnings surprises), or a combination of both. However, the widening confidence intervals at longer horizons indicate that this extended drift is noisier and less reliable at the individual event level.

Limitations

Several limitations should be considered when interpreting these results.

Look-ahead bias in universe filter. The market capitalization filter uses current (non-point-in-time) market caps. A stock that was below \$300M at the time of the event but has since grown could be incorrectly included, and vice versa. This may bias the sample toward ex-post survivors. A future extension using point-in-time market cap data from CRSP or Compustat would address this concern.

Data quality and survivorship. Price data from yfinance may have gaps due to ticker changes, delistings, or data provider limitations. Tickers that failed to download are logged, but their absence may create survivorship bias if firms that experienced negative outcomes after insider purchases are disproportionately missing. The strict NaN policy (any missing day in a window invalidates the event) mitigates partial-data bias but does not address tickers absent from the dataset entirely.

Filing date vs. trade date. The pre-registered event date is the filing date, not the trade date. Under SEC rules, Form 4 must be filed within two business days of the transaction, but late filings do occur. Using the filing date is conservative in the sense that it reflects the public information set, but it may also introduce noise if some filings in a cluster are late and the market has already reacted to the underlying trades. Some filings may occur after price adjustments have already begun, reducing measured post-event abnormal returns.

No transaction cost or capacity analysis. The reported abnormal returns are gross of transaction costs, slippage, and market impact. Whether the signal supports a profitable trading strategy depends on implementation costs, which are not modeled here.

Single benchmark. The primary analysis uses SPY as the sole benchmark. While the CAPM robustness check provides a risk-adjusted alternative, neither approach controls for size, value, momentum, or other factor exposures. A multi-factor model such as the Fama-French three- or five-factor specification could provide a more comprehensive adjustment for systematic risk exposures.

8. Conclusion

Clustered insider open-market purchases — defined as 3+ unique insiders filing purchases for the same stock within 30 calendar days — are associated with positive and statistically significant abnormal returns in U.S. equities over the 2015–2024 period. The primary pre-registered test shows a mean 30-day cumulative abnormal return of +1.65% ($t = 5.42$, $p = 6.6e-08$). The result is stable across time subperiods, robust to CAPM risk adjustment, and not sensitive to moderate changes in signal parameters.

As discussed in Section 7, these findings are consistent with the hypothesis that coordinated insider purchasing conveys stronger information than individual insider transactions. The per-event magnitude substantially exceeds the ~6% annual return estimated by Jeng et al. (2003) for general insider purchases,

supporting the interpretation that cluster events filter for high-conviction insider behavior. The pre-registered design ensures that the result is not an artifact of specification search.

Future work could extend the analysis in several directions: point-in-time market cap data to eliminate look-ahead bias, Fama-French factor adjustments for richer risk modeling, transaction cost modeling to assess implementability, and calendar-time portfolio construction to evaluate the signal as a continuous investment strategy. While the results suggest informational content in clustered insider purchases, implementation costs and real-time data constraints may affect practical profitability.

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